Glass Heroes v0.3.3.5 / Issue #31 — EXPERIMENTAL DEVELOPMENT REVIEW

Authorized customer/repair workflow and final menu-polish candidate. No release, merge or installation performed. GTA/Windows execution NOT RUN; Enhanced runtime compatibility NOT VERIFIED; historical crashes remain UNATTRIBUTED.

Baseline, owner evidence and branch

Work only on feature/shvdn3-335-customer-repair-workflow. Fetched exact remote branch at 1582ae8443704cc35926723f11dee8ef912d44be (authorization/handoff docs). Product source matched Issue30 branch commit 53205ad36aa398ebf6c9cb6626ea0d21ad135c0b, SHA256 4aba9e70bdc6c29a551cd4feb257c03d5bec7bf7218eedf7b32a1e3888a28e71. Read complete committed implementation handoff and public Issue31 body.

Owner explicitly confirmed using this approved repository baseline while the installed RC2 hash remains unverified, with no conflicting evidence/additional regressions reported. Did not infer deployed identity from a tag. RC1, RC2, Issue29/30, financial/WP1/FS1/integration worktrees, Issue28, main and QA R2 preserved. Original v2 code/project hashes unchanged. No new milestone beyond .5 begun.

Product source: experimental/shvdn3/Install/scripts/GlassHeroesBusiness.3.cs. Existing project directly links this source; net48/C#7.3, official ScriptHookVDotNet3 3.6.0. No dependency changes. Header/BOOT identify .5; SHA256, not reused labels, identifies exact code.

Focused implementation commits

- 5cf4321: visible Mechanic Break Area rename and role-based walking routes, existing point/mask handlers reused. Initial navigation 100/100 and Release 0 errors/two inherited warnings.

- 854c9d9: demonstrated finance/supplier Back-path corrections; 65-page menu audit. Focused navigation then 166/166.

- da14ab5: FIFO survival, usable bay eligibility, entity-creation retry backoff and destroyed staged vehicle validation. Initial focused workflow 16/16.

- 4eff687: assignment ownership and invalid-entity pre-settlement guard, including physical ON destroyed-car cleanup; focused workflow then 36/36.

- 8dfa965: existing on-demand customer status names mechanic/bay and clarifies wait; header aligned. Workflow 38/38.

- Final verification/documentation commit: exact SHA reported by branch log and completion response. Contains actual full-run evidence, expanded navigation, source-preservation and Word/PDF report; no additional product change.

Confirmed defects and fixes

1. Owner-reported break label and flat route discovery: corrected consistently in visible placement/editor/legend/help. `_breakArea`, EmployeeBreakArea and all XYZ/headings untouched. Walking Routes & Approaches now has Mechanics, Receptionist, Manager, Customers, Supplier Driver. Existing central hub still has six categories.

2. Financial Reports opened from Expenses ignored its return flag and went to Management. Back now restores Expenses row 8; direct Accounting still returns to Management row 3. No financial actions/math changed.

3. Pallet Storage entered from Supplier Settings fell back to Inventory. Capture its calling context so Back restores Supplier Settings row 4. Existing pallet buying/capacity/receipt logic unchanged. Correct stale overdue-wage notice

to Accounting > Expenses.

4. When staged Car 2 disappeared and Car 3 survived, the next visual update could spawn a newer Car 2, which dispatch preferred. Do not refill Car 2 while a surviving Car 3 exists; dispatch retains the older live reservation/vehicle. Existing promotion, pending-request decrement and one-drop-off checks remain.

5. Spawn failure set `_nextCustomerAt`, but pending dispatch did not consult that scheduling value, allowing creation attempts each visual second. Add session-only GTA-time backoff shared by direct/staged creation: 30s after creation failure, 120s after invalid direct model. Guard before spawning/model requests; retain requests and preserve safety/occupancy gates. Model streaming and ordinary obstruction are not redesigned. No thread, blocking wait or persisted retry timestamp.

6. Intake accepted a free worker without requiring a configured usable logical bay. CustomerBayAvailable now validates configured work/car target and existing order ownership; CanStartNextIntake uses it. Existing second intake during another repair remains permitted when drop-off, receptionist, free worker/bay and stock permit. Paid return priority/intake exclusivity unchanged.

7. Physical ON assignments did not protect coincident saved bay targets, and OFF settlement freed HasJob before its paid vehicle left the bay. Reuse CustomerBayAvailable in assignment: reserve worker slot and targets within conservative 4m until payment/return stages free them. Prefer existing specialists; keep original ON all-custom-car-bays-unset legacy fallback. This is logical ownership, not a new world scan or assignment framework.

8. Visual validation rejected destroyed vehicles only for OFF mode; economy can run before visual cleanup and physical settlement lacked a corresponding vehicle/customer guard. Reject missing/dead vehicle/customer before any work/completion credit and use existing cancellation/promise preservation. Paid retry protection, parts consumption, quote save and financial settlement remain unchanged.

Source-confirmed behavior is not proof these defects caused a particular owner's GTA symptom. No CLR or startup crash attribution follows from this work.

State transitions and unchanged behavior

FIFO: lost oldest entity -> existing validation cleans owned entities/retains request -> surviving third stays oldest -> no newer first-slot replacement -> existing activation consumes one request and claims drop-off. Temporary blockers retain pending work. Staged route timeouts remain bounded; no unnecessary recirculation or queued-entity recreation added.

Intake: existing active count, reserved stock, drop-off/return ownership, worker-at-break and receptionist gates remain. New logical bay gate only excludes unusable/reserved choices. Up to three active customer slots and two staged cars remain bounded. One/two/three staff tests exercise independent assignments; one staffed bay serializes safely.

Assignment: freeze/commit accepted quote before movement, rollback failed save and reserve exactly one worker/bay. ON retains walk/enter/drive/park/exit/return tasks and legacy fallback. OFF retains one-shot placement/heading, existing employee animation/work, settlement and pickup, including occupancy/player safety, partial-heading retry and bounded queries. No transfer helper or driving algorithm changed.

Completion: invalid vehicle/customer -> existing cancellation and unpaid promise preservation. Otherwise existing arrival/parking/work gates -> TrySettleRepair exactly once -> saved paid state/inventory/cash -> original invoice/notification -> clear pickup or retained paid wait. No second repair from blocked pickup. Cancellation logic and native entity cleanup ownership unchanged. Invoice CSV remains inherited best-effort logging; FinanceV1/committed state is authoritative if CSV writing fails.

Missing worker recovery, physical movement timeouts and route recovery were audited and exercised through deterministic state adapters. No speculative new driving recovery, forced occupied-car teleport or persistent completed-car serialization was added. A deleted worker can still defer while original respawn/route policy applies; actual GTA task recovery remains manual acceptance.

Before:
```
CustomerOrder next = _stagedCustomerOrder ?? _stagedThirdCustomerOrder;
```

```
// A failed Car 2 could be refilled before this existing selection.
```

After: existing selection is unchanged; SpawnRoadsideWaitingCustomer refuses a new first slot while a surviving third reservation exists.

Before: CanStartNextIntake only required a free at-break worker; assignment only checked HasJob/configuration. After: both consult bounded CustomerBayAvailable, with per-order ownership and saved-target overlap checks.

Before: physical destroyed cars could pass visual/settlement gates. After:

```
if (order.Vehicle == null || !order.Vehicle.Exists() || order.Vehicle.IsDead ||
    order.Customer == null || !order.Customer.Exists() || order.Customer.IsDead)
{ CancelCustomerOrder("Customer entity unavailable before repair settlement", true); return; }
```

Full source.diff is against the exact approved Issue30 product and is verified to reconstruct the final source.

Menu organization and complete audit

Mechanics role links: existing five arrival points; five SHARED work points; three bay approaches; MechanicRouteRoster and each worker's outbound/return masks; legacy checkpoint; existing Walking Mode & Recovery page. It does not create fifteen independent physical points. Receptionist/manager link their existing stops, final approach and mode pages. Customer/supplier roles link their existing stops and approach/handoff. PointEditor uses the same walk/facing capture, restrictions, saved status and coordinates/heading view.

Existing `_placementMenuReturns` provides nested page/selection return. Native keyboard/controller handlers, debouncing, phone/input suppression, menu rendering/scrolling, native Esc/Start pause and actual point editor remain unchanged. Added enum pages are appended. Shared modes/settings are links, not duplicated state. Operational supplier/customer/employee controls remain where expected.

NAVIGATION_MATRIX.md lists ALL 65 pages, labels/titles, default Back target/row and contextual parent overrides. The Word/PDF appendix includes the matrix. It distinguishes neutral-context shape/Back tests from contextual role/editor/controller tests. Per-page tests do not automatically select purchase/destructive actions; inherited action suites and source review cover unchanged transactional handlers. Actual hardware, scrolling visibility and safe controller walking/look need owner GTA validation.

Modified methods and preservation evidence

Menu methods: GetMenuItems, GetMenuItemValue, GetMenuTitle, GetSelectedDescription, ExecutePlacementSelection, ExecuteSelection, NavigateBack, ExecuteEmployeePlacementsSelection, ExecuteMarkerSelection, ExecuteSupplierDeliverySetup.

Workflow/status: CanStartNextIntake, AssignCustomerMechanic, SpawnCustomerOrder, SpawnRoadsideWaitingCustomer, ValidateStagedCustomer, UpdateOneCustomerOrder, ProcessOneCustomerRepair, ShowCustomerOrderStatus. Constructor BOOT label only; ProcessEconomy notification string only. New helpers: CustomerBayAvailable, CustomerSpawnRetryReady, DeferCustomerSpawn. Session-only retry fields; no entity fields persisted.

Preservation pins 346 existing methods unchanged. Additional exact assertions prove constructor label-only and economy notification-only deltas. Freeze Save, FinanceV1 load/write, TryCommitBusinessSave, TrySettleRepair, RecordCustomerInvoice, OnTick/OnAborted, all point capture/load/save/render helpers, remote operations, supplier order/receipt, cancellation, driving and one-shot teleport helpers. Existing approximately one-second customer visual cadence unchanged; new ownership loops bounded by three slots. No new world enumeration, expensive Tick work, spawned type, thread or recursive callback.

Save.ini/Settings.ini schema changes: NONE. No location/key/default reset; EmployeeBreakArea unchanged. Finance, inventory, mixed supplier orders, manual retail, wages, pallets, staff XP and saved accepted quotes protected by preservation and inherited tests. Source fixtures are verification inputs only, not install scripts.

Actual validation and limits

Final exact-source full run passed 671/671 (467 inherited + 166 menu + 38 workflow), then the navigation suite added all three mechanics x two directions x DIRECT/CUSTOM configuration variants and reran 178/178 with NO further product changes. Combined total is 683/683. The log/validation records these actual runs; no redundant second full run is claimed after test-only expansion.

Focused workflow compiles production intake/staging/spawn/assignment/order-state/cancel/settlement logic against deterministic native/entity/movement adapters and temporary-file quote saves. It tests physical timeout state dispatch, not GTA driving AI. Existing 34 bay tests also exercise actual one-shot placement logic through native adapters; 41 quote tests cover save failure, abort, restart and no offline progress. New menu serializers roundtrip 59 placement sections with flags/XYZ/headings and unrelated cash sentinel, through a ScriptSettings file adapter. All real serializers are additionally source-pinned.

The actual Release project compiles net48/C#7.3 against SHVDN3 3.6.0 using fresh temporary obj/output. Warnings are inherited CS0169 (_selectedInvoiceMechanicSlot unused) and CS0649 (CustomerOrder.VehicleTaskIssued never assigned). This is not engine testing and does not certify Enhanced 3.9.0.6, LS RED interoperability or stability.

Reproduce:

```
source /workspace/.cloud-setup/env.sh
python verification/workflow335/verify.py --label v0.3.3.5-issue31 --profile workflow335
--suites repair sales pallets wages reload supplier range menu cart pause focus finance startup
hardening bays workflow335
```

Known limitations, hypotheses and next test

- Paid departure/pickup vehicles and unaccepted queue requests remain session transient, as in the baseline. Reload can clean a paid car while preserving its paid cash state. No persistent completed-car design was introduced.

- Logical saved-target spacing uses conservative 4m clearance; dense MLO bays may defer. Physical collision/pathfinding remains engine-dependent. Misconfigured points still need correction.

- A failed oldest staged entity cannot preserve that physical customer's identity; its pending request survives, and the surviving older live reservation gets priority over a replacement. No durable unaccepted queue identity added.

- Creation cooldown uses session GTA game time and paused/boot guards, not wall-clock offline recovery. Existing model-request/streaming behavior otherwise remains; a model that never loads is not automatically substituted.

- Physical lost-worker/parking/enter/exit recovery and manager/streaming transitions retain original policy. Source audits/tests do not prove repeated idle simulation failures or historic crashes resolved.

- Some original operational shortcuts remain intentionally; all roles share the same existing editors. Treat new owner menu reports as .5 stabilization issues before a larger milestone, not proof that all runtime UI defects are eliminated.

Next test: owner source-verified GH + LS RED extended session using GTA_TEST_CHECKLIST.md, first all role/edit/Back/controller paths, then parallel queue/repair ON/OFF and save/reload. Record IDs, source/runtime hash and exact paths for defects. Owner chooses duration; no invented stability threshold.

GTA checklist, rollback and package readiness

GTA_TEST_CHECKLIST.md is the detailed check sheet, including every menu/role, each worker/direction, one/two/three staff and queues, FIFO loss case, native input, pause/restart, transactions and extended observation. Restore COMPLETE matching source/data/journal backups for rollback; never alternate independent builds against modified sole Save.ini. No live installation accessed.

After code review and explicit release approval only, a proposed v0.3.3.5-rc1 package can contain this exact single .3.cs, matching optional project, checksum/manifest, installer-free guides and reports. No archive, tag, release or

installation is created by this milestone. No main/QA/Issue28 merge or next feature work.

Recorded automated results

Genuine SHVDN3 3.6.0 net48 / C#7.3 Release: PASS, 0 errors; inherited warnings CS0169 and CS0649. SDK 8.0.426.

683/683 behavior tests PASS: 85 inherited issue/logger/migration checks plus repair 32, sales 29, pallets 25, wages 22, reload 41, supplier 37, range 40, menu 13, cart 32, pause 9, focus 8, finance 33, startup 18, hardening 9, bays 34, workflow335 38; navigation 178. Static 21/21 PASS; focused scope/save/v2/clock/settlement/transfer-helper preservation PASS. The new navigation tests use production members/case arms with deterministic adapters, not GTA.

Candidate source SHA256: 741c97730856adad11cd98d985e41d4eb325c9788af5ceebeee965a0eb64125f. Full diff verified to reproduce this source from 53205ad.

GTA testing NOT RUN. Windows execution NOT RUN. Enhanced runtime compatibility NOT VERIFIED. Crash cause UNATTRIBUTED. No release archive, runtime deployment, save modification or merge was produced by this task.

v0.3.3.5 / Issue #31 — owner GTA check sheet — NOT RUN

Target: GTA V Legacy 1.0.3889.0, SHVDN Enhanced 3.9.0.6, Glass Heroes + Los Santos RED first. Compilation against official SHVDN3 3.6.0 does not certify Enhanced compatibility. Owner determines extended-session duration; this is not a fixed stability test or Issue28 resolution.

Before a future authorized test installation

- [ ] Review this candidate first; no release or installation is performed by this task. Preserve the exact working RC2 script and COMPLETE matching scripts/GlassHeroesBusiness directory outside scripts (Save.ini, Settings.ini, journals, CSV, backups/recovery files and logs). Keep a sole-original backup untouched and use disposable QA copies.

- [ ] Record installed source SHA256, GTA/runtime/mod versions and selected transfer/walking/parking modes. Exactly one Business Manager .3.cs or compiled equivalent, never both. Default 30-second active-game startup gate; confirm one instance, correct .5 BOOT/source hash and no initialization/save errors.

- [ ] Compare all preexisting cash, financial records, stock, pending suppliers/quotes, staff progression/assignments, points and headings before any edit. Do not alternate candidate source on modified data without restoring its complete matching backup.

Menus and placement

- [ ] Use NAVIGATION_MATRIX.md to visit all 65 pages and every safe view row. Exercise keyboard and controller, Up/Down wrapping/scroll, Accept once/held-repeat/frontend echo, visible Back and Backspace/controller Back, close/reopen F2, Esc and Start native pause. Walk/look safely with controller during point placement; watch for input suppression, clipping, wrong action or stale parent.

- [ ] Management has only Accounting > Expenses & Bills. Reports opened from Expenses return to Expenses row 8. Equipment/invoices/wage/supplier settlement remain accessible; browsing never purchases/pays/writes unexpectedly. Owner deposit/withdraw and controller/shop rows map correctly; do not move shop just to browse.

- [ ] Placement Settings has six categories. Walking Routes has Mechanics, Receptionist, Manager, Customers and Supplier Driver. Check all role Back rows and exact highlighted parent selections. Old Customers/Employees hub shortcuts still return to their original rows.

- [ ] Employee Locations says Mechanic Break Area; editor and legend agree. Existing position/heading survives save/reload under EmployeeBreakArea. Do not reset it. Receptionist and manager destinations remain separate.

- [ ] Mechanics: all five arrival and five shared work stops, three final bay approaches, legacy checkpoint, existing Walking Mode & Recovery link. Job Waypoint Selection: Mechanics 1/2/3 each outbound and return; toggling one mask changes only that mechanic/direction. There are FIVE shared physical work points, not fifteen independent coordinates. Test DIRECT/CUSTOM modes without editing unrelated points.

- [ ] Receptionist/manager roles: five stops, final approach, existing walking-mode link. Customers/supplier roles: three stops and reception approach/handoff. Save Position + Facing on disposable data, Show Saved Coordinates, Back to exact list/role/row, reload. No role edits another role's route.

- [ ] Service Bays 1/2/3: separate mechanic animation vs car coordinates/headings, fixed assignments, AUTO LOT/CUSTOM. Existing closed-shop/active-order editing restrictions remain. Marker display ON/OFF changes visibility only. Existing AUTO LOT preview independent.

- [ ] Delivery & Stockroom: supplier spawn/unload/handoff, owned pallets and driver stops. Pallets opened from Supplier Settings return there, not Inventory. Equipment from hub returns there; equipment from Expenses returns to Expenses. Purchases remain explicit existing actions.

Queue, assignments and vehicles

- [ ] One customer/one mechanic: full intake, accepted quote, work arrival/animation, one payment/parts consumption, pickup and departure. Repeat with two then three mechanics, all staffed usable bays configured.

- [ ] Three queued customers/one staffed bay: roadside Cars 2/3 survive while first is serviced; enter FIFO. With three mechanics, Customer 2 checks in while Customer 1 still repairs, only when drop-off/receptionist/another usable staffed bay are free. Each worker owns one order; distinct saved targets must be clear.

- [ ] Obstruct drop-off/pickup and coincident/occupied bay targets: no double-booking, head-on return or premature check-in. Returning paid car gets priority; clearing obstruction resumes. Conservative 4m logical bay separation may defer tightly spaced MLO bays; report actual coordinates/mode.

- [ ] Transfers ON: walk to car, enter, drive/park, exit, work animation, return drive/customer pickup. Temporarily impede safe entry/exit/parking on disposable QA; existing timeouts cancel/defer without occupied-car teleport or continuous tasks. Misconfigured MLO points require owner correction, not unsafe forced movement.

- [ ] Transfers OFF: one safe bay placement and one safe pickup placement; no occupied car/player teleport, no repeated position write. Block pickup after payment: retain paid state, no second repair/invoice; resume once free.

- [ ] On disposable data, remove/destroy queued/active vehicles or lose a mechanic/customer: safe cancellation/defer and accepted quote retained where applicable, no ghost payment. If staged Car 2 disappears but Car 3 survives, no newer replacement jumps ahead. Preserve request count.

- [ ] If entity creation fails naturally, observe retained request and bounded retries (30 active-game seconds; invalid direct model 120 seconds). Do not deliberately corrupt the main setup. Timeouts already bound route failure; new cooldown is only for spawn/model failure, not endless fresh customer creation.

- [ ] Use Customer Status on demand: assigned mechanic/bay, other live order stages, waiting state; no per-frame spam. Record IDs/stages, quote and before/after cash/stock/invoice for each unexpected condition.

Persistence, economy and extended session

- [ ] Reload/abort during check-in, accepted assignment and repair with COMPLETE disposable matching data. Accepted quotes/remaining work recover; no old entity handles or stale movement command. Failed-save settlement must remain unpaid and retry once after save access recovers; do not damage a live save to test failure injection.

- [ ] Reload during paid blocked pickup: no duplicate payment. Paid session cars remain transient and may be cleaned up, as in Issue29; completed-car serialization is NOT implemented. Unaccepted roadside requests/entities are also session transient; do not infer durable order acceptance before a quote save.

- [ ] Pause pending work/ETA, wait, resume: no paused accrual. Close GTA and restart: no offline catch-up. Regress remote manager jobs, $5,000 revenue reconciliation with/without settled expenses, unpaid wage/supplier accrual versus actual cash payments, retail sales, mixed supply orders, wholesale player-only and nine-box pallets.

- [ ] Conduct the owner's longer GH + LS RED session. Record duration and exact menu selection path for each defect; capture SHVDN/RPH/Errors logs and dump with source hash. Keep gameplay CLR overflow, startup AV and EUP/file-access findings separate. Additional menu reports stay in this .5 stabilization milestone for triage before larger feature development.

Rollback

Close GTA; preserve the entire tested source/data/log set separately. Remove .5 and restore the exact prior single RC2 script AND its COMPLETE matching data directory. Do not restore only Save.ini over another journal or leave duplicate scripts. Full rollback discards test-session progress; preserve the untouched original backup. No rollback/deployment command was executed against a live installation by Codex.

v0.3.3.5 — complete F2 navigation audit matrix

All 65 MenuPage values are listed below with production menu labels, title, and default Back page/selection from the deterministic harness. Default Back is exercised with neutral context fields; role/hub/editor/shortcut parents depend on captured entry context and are separately tested by role/editor roundtrip cases. A row here does not claim GTA execution or automatic selection of every purchase/destructive action. Payment/action algorithms are covered by inherited suites; source routes were reviewed for each handler.

65 per-page cases check nonempty titles/labels, keyboard wrapping and default Back without browsing writes. Focused cases also execute keyboard/controller accept/Back and nested role context. Actual controller hardware, scroll rendering and GTA movement remain manual acceptance.

| Menu page | Title | Default Back target / selected row | Items |

| --- | --- | --- | --- |

| Business | GLASS HEROES | Business / 0 | Open Business; Save Business; Business Status; Business Management > |

| BusinessManagement | BUSINESS MANAGEMENT | Business / 3 | Employees >; Customers >; Inventory & Suppliers >; Finances & Accounting >; Placement Settings >; Controller Controls >; Deposit $1,000; Withdraw $1,000; Set Shop Location; < Back to Main |

| Employees | EMPLOYEES | BusinessManagement / 0 | Employee Status; Mechanics >; Receptionist >; Business Manager >; Employee Settings >; < Back to Business Management |

| EmployeeSettings | EMPLOYEE SETTINGS | Employees / 4 | Placement Settings >; Employee Walking Paths >; Blocked Path Recovery >; Placement Markers >; Break-Time Animations; < Back to Employees |

| EmployeePaths | EMPLOYEE WALKING PATHS | EmployeeSettings / 1 | Mechanic Walking Paths >; Receptionist Walking Paths >; Manager Walking Paths >; < Back |

| ArrivalWaypoints | SPAWN -> BREAK (0-5) | MechanicWalkingPaths / 2 | Arrival Point 1 / 5 >; Arrival Point 2 / 5 >; Arrival Point 3 / 5 >; Arrival Point 4 / 5 >; Arrival Point 5 / 5 >; < Back |

| MechanicBays | SERVICE BAYS | EmployeePlacements / 1 | Service Bay 1 >; Service Bay 2 >; Service Bay 3 >; < Back |

| ServiceWaypoints | BREAK -> BAY (0-5) | MechanicWalkingPaths / 3 | Work Route 1 / 5 >; Work Route 2 / 5 >; Work Route 3 / 5 >; Work Route 4 / 5 >; Work Route 5 / 5 >; < Back |

| BayApproaches | BAY APPROACHES | MechanicWalkingPaths / 4 | Bay 1 Final Turn >; Bay 2 Final Turn >; Bay 3 Final Turn >; < Back |

| ServiceVehicleBays | SERVICE VEHICLE PARKING | CustomerPlacements / 6 | Service Bay 1 >; Service Bay 2 >; Service Bay 3 >; < Back |

| ServiceBayProfile | SERVICE BAY 1 | Business / 0 | Set Vehicle Position (CUSTOM); Set Mechanic Work Position; Mechanic Assignment; View Current Configuration; < Back to Service Bays |

| PointEditor | TEST POINT | Business / 0 | Set / Save Position + Facing; Clear Saved Position; Marker Preview; Show Saved Coordinates; < Back |

| RouteRecovery | ROUTE RECOVERY | EmployeeSettings / 2 | Blocked-Route Teleport; Recovery Status; < Back |

| PlacementMarkers | PLACEMENT MARKERS | EmployeeSettings / 3 | Placement Markers; Show Marker Color Legend; < Back |

| Customers | CUSTOMER SERVICE | BusinessManagement / 1 | Customer Order Status; Customer Queue >; Placement Settings >; Vehicle Driving Options >; Customer Arrivals; Schedule Next Customer; Customer Blips; Placement Markers; < Back to Business Management |

| Finances | FINANCES & ACCOUNTING | BusinessManagement / 3 | Financial Overview >; Income & Sales >; Expenses & Bills >; Profit & Loss >; Transaction History >; Owner Cash Transfers >; Reporting Period: Toggle; < Back to Management |

| FinancialOverview | FINANCIAL OVERVIEW | Finances / 0 | Reporting Period: Toggle; Period Closing Cash; Gross Revenue; Cash Flow Reconciliation; Outstanding Bills at Close; Save / Audit State; < Back to Accounting |

| IncomeSales | INCOME & SALES | Finances / 1 | Reporting Period: Toggle; Physical Customer Repairs; Manager Remote Repairs; Autonomous Mechanic Jobs; Parts Retail Sales; Gross Revenue (No Overlap); Known Rebates / Refunds; < Back to Accounting |

| ProfitLoss | PROFIT & LOSS | Finances / 3 | Reporting Period: Toggle; Gross Revenue; Cash Expense Payments; Expenses Incurred; Accounting Profit; Accounting Basis / Notes; < Back to Accounting |

| TransactionHistory | TRANSACTION HISTORY | Finances / 4 | #1 test history; Older Transactions >; Newer Transactions >; < Back to Accounting |

| OwnerTransfers | OWNER CASH TRANSFERS | Finances / 5 | Deposit Owner Funds; Withdraw Owner Funds; Recorded Owner Deposits; Recorded Owner Withdrawals; Reporting Period: Toggle; < Back to Accounting |

| StaffRoster | MECHANIC ROSTER | MechanicManagement / 0 | Mechanic 1 >; Mechanic 2 >; Mechanic 3 >; < Back to Mechanics |

| StaffProfile | MECHANIC PROFILE | StaffRoster / 0 | Profile / Performance; Training: +40 XP; Change Specialization; Raise Period Wage (+$50); Fire This Mechanic; < Back to Roster |

| VehicleDriveOptions | VEHICLE DRIVING | Customers / 3 | Drive-in Customer Cars; Mechanic Vehicle Transfers; Service Parking Mode; Auto Lot Settings >; Custom MLO Parking Bays >; Reverse Before Driving; Driving Status / Diagnostics; < Back to Customers |

| CustomerPlacements | CUSTOMER PLACEMENTS | Customers / 2 | Customer Drop-off Parking >; Customer Road Spawn >; Customer Reception Approach >; Customer Check-in Position >; Waiting Car 2 Position >; Waiting Car 3 Position >; Service Vehicle Parking Bays >; Customer Walking Route (0-3) >; < Back |

| EmployeePlacements | EMPLOYEE PLACEMENTS | EmployeeSettings / 0 | Mechanic Break Area >; Service Bays >; Receptionist Desk >; Receptionist Desk Approach >; Manager Workstation >; Manager Workstation Approach >; Staff Spawn / Entry >; < Back |

| MechanicWalkingPaths | MECHANIC WALKING | EmployeePaths / 0 | Mechanic Walking Mode; Pickup + Return Exit Route; Spawn -> Break Waypoints >; Break -> Bay Waypoints >; Final Bay Approaches >; Legacy Staff Checkpoint (MLO) >; Choose Job Waypoints by Mechanic >; Smooth Through-Waypoints; Teleport Stuck Pedestrians; Advanced Break <-> Bay Routes; < Back |

| ReceptionWalkingPaths | RECEPTION WALKING | EmployeePaths / 1 | Receptionist Walking Mode; Spawn -> Desk Waypoints >; Final Desk Approach >; < Back |

| ReceptionWaypoints | RECEPTION ROUTE (0-5) | ReceptionWalkingPaths / 1 | Reception Stop 1 / 5 >; Reception Stop 2 / 5 >; Reception Stop 3 / 5 >; Reception Stop 4 / 5 >; Reception Stop 5 / 5 >; < Back |

| CustomerQueueSettings | CUSTOMER QUEUE | Customers / 1 | Pending Customer Requests; Maximum Queue Capacity; Schedule One Customer; Clear Pending Requests; < Back to Customers |

| PartsSales | PARTS SALES - ONE UNIT | InventoryManagement / 3 | Universal Part; Service Filter; Brake Kit; Diagnostic Supply; < Back to Inventory |

| VehicleAutoLotOptions | AUTO LOT PARKING | VehicleDriveOptions / 3 | Car Parking Side; Car Offset from Mechanic Bay; Auto Parking Preview; < Back to Driving |

| BusinessExpenses | EXPENSES & BILLS | Finances / 2 | Outstanding Bills; Mechanic Wage Invoices >; Pay Receptionist Wages; Pay Rent & Utilities; Pay Supplier Invoices; Pay All Bills; Order 10 Universal Parts; Equipment & Office >; Financial Reports >; Pay Manager Wages; < Back to Business Management |

| MechanicInvoices | MECHANIC WAGES | BusinessExpenses / 1 | Mechanic 1 Wage Invoice; Mechanic 2 Wage Invoice; Mechanic 3 Wage Invoice; Pay Former Employee Wages; Pay All Mechanic Wages; < Back to Expenses |

| CustomerWalkingRoutes | CUSTOMER WALK (0-3) | CustomerPlacements / 7 | Customer Walking Stop 1 / 3 >; Customer Walking Stop 2 / 3 >; Customer Walking Stop 3 / 3 >; < Back |

| InventoryManagement | INVENTORY & SUPPLIERS | BusinessManagement / 2 | Inventory Overview >; Build Supplier Order >; Active Deliveries >; Parts Sales >; Stockroom & Pallets >; Supplier Settings >; < Back to Management |

| InventoryOverview | INVENTORY OVERVIEW | InventoryManagement / 0 | Universal Parts; Service Filters; Brake Kits; Diagnostic Supplies; Storage Capacity; Reorder Advice; Low-stock Threshold; < Back to Inventory |

| BuildSupplierOrder | BUILD SUPPLIER ORDER | InventoryManagement / 1 | Universal Parts; Service Filters; Brake Kits; Diagnostic Supplies; Supplier: Change; Cost / Capacity Preview; Place Purchase Order; Fill Low-stock Cart; Clear Cart; < Back to Inventory |

| ActiveDeliveries | ACTIVE DELIVERIES | InventoryManagement / 2 | Delivery Status; Order Contents; Recover Original Order; Cancel Pending Order; < Back to Inventory |

| EquipmentManagement | SHOP EQUIPMENT | BusinessExpenses / 7 | Equipment Status; Buy Tool Set; Buy Diagnostic Scanner; Buy Desk + Laptop; Place Tool Chest; Remove Tool Chest; Place Diagnostic Trolley; Remove Diagnostic Trolley; Place Desk in Front; Remove Placed Desk; Office PC Status; < Back |

| ControllerSettings | CONTROLLER INPUT | BusinessManagement / 5 | Controller Menu Controls (ENTER); View Controller Key Map; Controller Input Diagnostics; < Back to Business Management |

| MechanicRouteRoster | MECHANIC JOB ROUTES | MechanicWalkingPaths / 6 | Mechanic 1 Job Route >; Mechanic 2 Job Route >; Mechanic 3 Job Route >; < Back to Mechanic Paths |

| MechanicRouteDirection | SELECT WALK DIRECTION | MechanicRouteRoster / 0 | Break -> Job Waypoint Selection >; Job -> Break Waypoint Selection >; Route Summary; < Back to Mechanic List |

| MechanicRoutePoints | JOB WALK WAYPOINTS | MechanicRouteDirection / 0 | Use Work Waypoint 1; Use Work Waypoint 2; Use Work Waypoint 3; Use Work Waypoint 4; Use Work Waypoint 5; All Waypoints ON; All Waypoints OFF; < Back to Route Direction |

| SupplierDeliverySetup | SUPPLIER SETTINGS | InventoryManagement / 5 | Physical Mule Delivery; Mule Road Spawn >; Truck Unloading Spot >; Driver Box Handoff >; Pallet Storage & Capacity >; Driver Walk Stops (0-3) >; Wholesale Warehouse: Fixed; Assign Current Pickup Vehicle; Assigned Vehicle / Cargo Status; Delivery Route / Stock Status; Recover Stuck Shipment; Cancel Pending Order (press twice); < Back to Inventory |

| SupplierDriverWaypoints | DRIVER WALK (0-3) | SupplierDeliverySetup / 5 | Driver Stop 1 / 3 >; Driver Stop 2 / 3 >; Driver Stop 3 / 3 >; < Back |

| HireMechanicRoster | HIRE A MECHANIC | MechanicManagement / 1 | Junior General Mechanic; Skilled Brake Technician; Experienced Diagnostic Technician; < Back to Mechanics |

| HireReceptionistRoster | HIRE RECEPTIONIST | ReceptionistManagement / 1 | Entry Receptionist (1 queue); Experienced Receptionist (2 queue); Senior Service Advisor (3 queue); < Back to Receptionist |

| MechanicManagement | MECHANIC MANAGEMENT | Employees / 1 | Mechanic Roster >; Hire Mechanic >; Mechanic Job Waypoints >; Mechanic Walking Paths >; < Back to Employees |

| ReceptionistManagement | RECEPTION MANAGEMENT | Employees / 2 | Receptionist Profile >; Hire Receptionist >; Receptionist Walking Paths >; < Back to Employees |

| ReceptionistProfile | RECEPTIONIST PROFILE | ReceptionistManagement / 0 | Profile & Queue Skill; Train Receptionist (+1 Capacity); Increase Period Wage (+$25); Dismiss Receptionist; < Back to Receptionists |

| SupplierPalletStorage | STOCKROOM & PALLETS | InventoryManagement / 4 | Storage Capacity Status; Purchase Additional Pallet ($750); Place / Move Pallet 1 >; Place / Move Pallet 2 >; Place / Move Pallet 3 >; Place / Move Pallet 4 >; < Back |

| ManagerManagement | BUSINESS MANAGER | Employees / 3 | Manager Profile >; Hire Business Manager >; Manager Walking Paths >; < Back to Employees |

| HireManagerRoster | MANAGEMENT CANDIDATES | ManagerManagement / 1 | Assistant Manager — 3% discount; Experienced Manager — 6% discount; General Manager — 10% discount; < Back to Management |

| ManagerProfile | MANAGER PROFILE | ManagerManagement / 0 | Profile & Business Reports; Train Manager (+40 XP); Pay Manager Wage Invoice; Dismiss Manager; < Back to Manager |

| ManagerWalkingPaths | MANAGER WALKING | EmployeePaths / 2 | Manager Walking Mode; Spawn -> Workstation Waypoints >; Final Workstation Approach >; < Back |

| ManagerWaypoints | MANAGER ROUTE (0-5) | ManagerWalkingPaths / 1 | Manager Stop 1 / 5 >; Manager Stop 2 / 5 >; Manager Stop 3 / 5 >; Manager Stop 4 / 5 >; Manager Stop 5 / 5 >; < Back |

| PlacementSettings | PLACEMENT SETTINGS | PlacementSettings / 0 | Customer Locations >; Employee Locations >; Service Bays >; Delivery & Stockroom >; Walking Routes & Approaches >; Marker Display Settings >; < Back to Management |

| DeliveryPlacements | DELIVERY & STOCKROOM LOCATIONS | DeliveryPlacements / 0 | Mule Road Spawn >; Truck Unloading Spot >; Driver Box Handoff >; Pallet Storage & Placement >; Supplier Driver Stops >; Equipment Placement & Upgrades >; < Back to Placement Settings |

| PlacementWalkingRoutes | WALKING ROUTES & APPROACHES | PlacementWalkingRoutes / 0 | Mechanics >; Receptionist >; Manager >; Customers >; Supplier Driver >; < Back to Placement Settings |

| MechanicPlacementRoutes | MECHANIC WALKING ROUTES | MechanicPlacementRoutes / 0 | Spawn -> Mechanic Break Area >; Mechanic Break -> Service Bays >; Final Bay Approaches >; Job Waypoint Selection by Mechanic >; Legacy Mechanic Checkpoint >; Walking Mode & Recovery >; < Back to Roles |

| ReceptionPlacementRoutes | RECEPTIONIST WALKING ROUTES | ReceptionPlacementRoutes / 0 | Spawn -> Desk Stops >; Final Desk Approach >; Walking Mode >; < Back to Roles |

| ManagerPlacementRoutes | MANAGER WALKING ROUTES | ManagerPlacementRoutes / 0 | Spawn -> Workstation Stops >; Final Workstation Approach >; Walking Mode >; < Back to Roles |

| CustomerPlacementRoutes | CUSTOMER WALKING ROUTES | CustomerPlacementRoutes / 0 | Customer Walking Stops >; Reception Approach >; < Back to Roles |

| SupplierPlacementRoutes | SUPPLIER DRIVER WALKING ROUTES | SupplierPlacementRoutes / 0 | Supplier Driver Stops >; Driver Box Handoff >; < Back to Roles |

Contextual parent overrides

- PlacementSettings returns to its actual Management/Customers/EmployeeSettings caller and row.

- Walking Routes returns to hub row 4; each of five role pages returns to its matching role row.

- Mechanic arrival/service/bay lists, route roster, checkpoint and mode shortcut return through Mechanics. Job mask pages return direction -> staff -> roster -> role.

- Receptionist/manager/customer/supplier stops and approach editors return to their role page/row.

- PointEditor always returns to its captured immediate parent/row, with no saved-coordinate rewrite.

- Shared Service Bay profiles return to the entry list and bay slot.

- Equipment and pallet pages return to the caller when entered through the hub. Pallets from Supplier Settings now return to Supplier Settings row 4.

- Reports from Expenses now return to Expenses row 8; direct Accounting returns to Management row 3.

No new input suppression, persistent overlay, point engine or saved menu state introduced. Existing controller debounce, native pause and phone suppression are pinned unchanged.